Understanding Airbnb Market Demand and Listing Segments in California

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Background

- Airbnb markets vary significantly across cities and neighborhoods
- Understanding what drives listing demand and how listings cluster into market segments can help hosts and platforms optimize pricing and strategy

Dataset Overview

A periodic snapshot of the active Airbnb market, capturing publicly available listing data to analyze supply, demand, and performance metrics.

Source

Inside Airbnb

Independent, non-commercial open data project

SNAPSHOT METADATA

Sept 2025	81,587	78
SNAPSHOT DATE	TOTAL LISTINGS	VARIABLES

GEOGRAPHIC COVERAGE (8 MARKETS)

- Los Angeles
- San Francisco
- San Diego
- Oakland
- Santa Clara County
- San Mateo County
- Santa Cruz County
- Pacific Grove

Data Limitations

Represents publicly visible active listings only. Does not include hidden, inactive, or brand new listings without public footprint. Booking data is estimated based on reviews and availability calendars.

METHODOLOGY

Data Cleaning Pipeline

airbnb

Processing Steps

1. **Combine Datasets**
   Merged 8 individual market files into a single dataframe, adding a 'location' column to preserve market identity.

2. **Feature Selection**
   Retained critical analytical columns: host attributes, property details, pricing, availability metrics, and review scores.

3. **Handling Missing Values**
   Created 'has_reviews' flag. Imputed missing review scores with 0 to accurately represent listings with no reviews.

4. **Price Cleaning**
   Stripped currency symbols ($) and commas. Converted to numeric format for statistical analysis.

5. **Outlier Removal**
   Removed listings in the top 1% of price distribution to prevent skewing of demand models.

Quality Checks & Impact

664
TOTAL ROWS REMOVED

~0.8%
DATA REDUCTION

Minimal data loss ensures statistical integrity while removing erroneous entries.

Market Balance Preserved

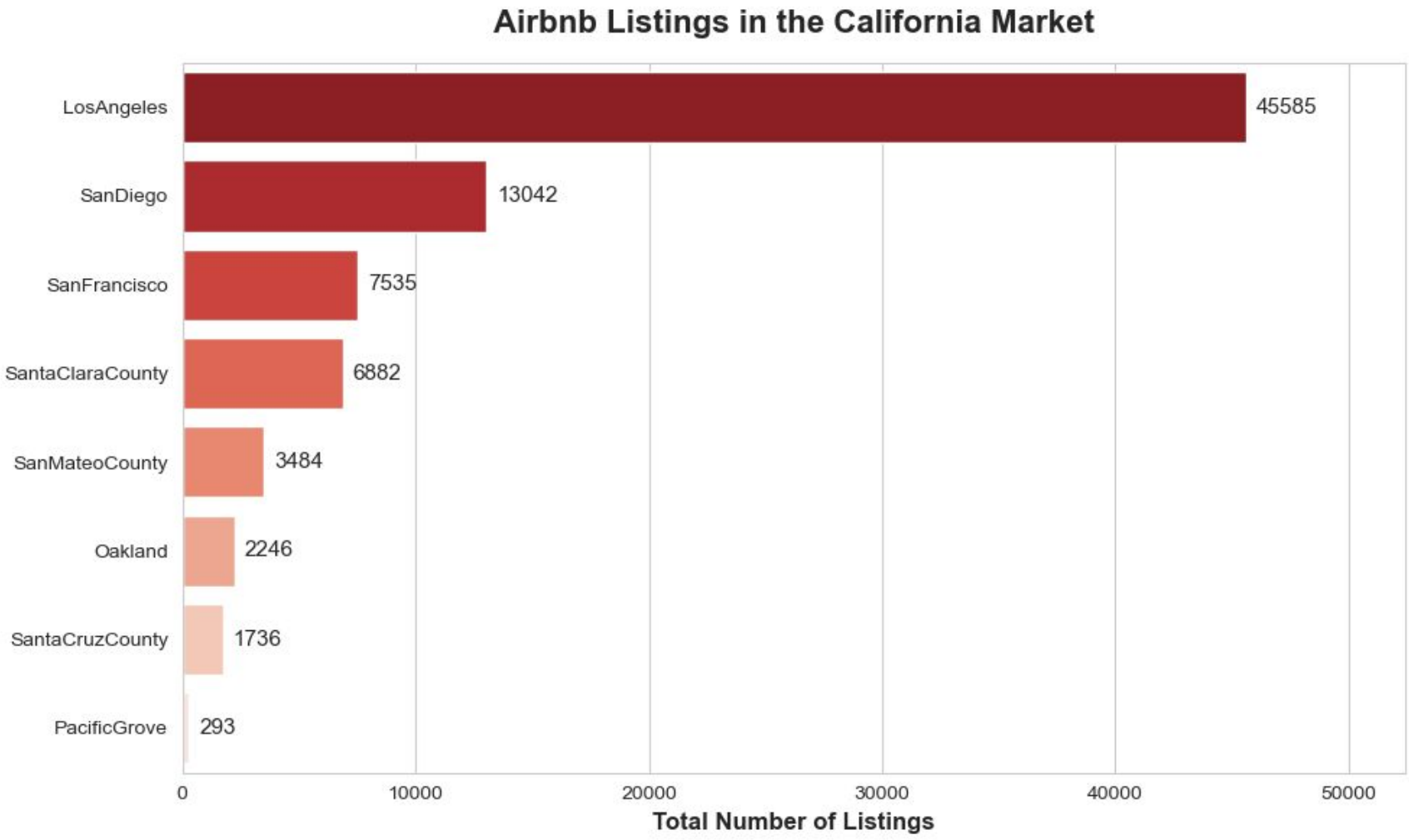
Cleaning impact was distributed proportionally across markets:

- Los Angeles: -374 rows
- San Francisco: -32 rows
- San Mateo: -32 rows
- San Diego: -135 rows
- Santa Clara: -61 rows
- Santa Cruz: -25 rows

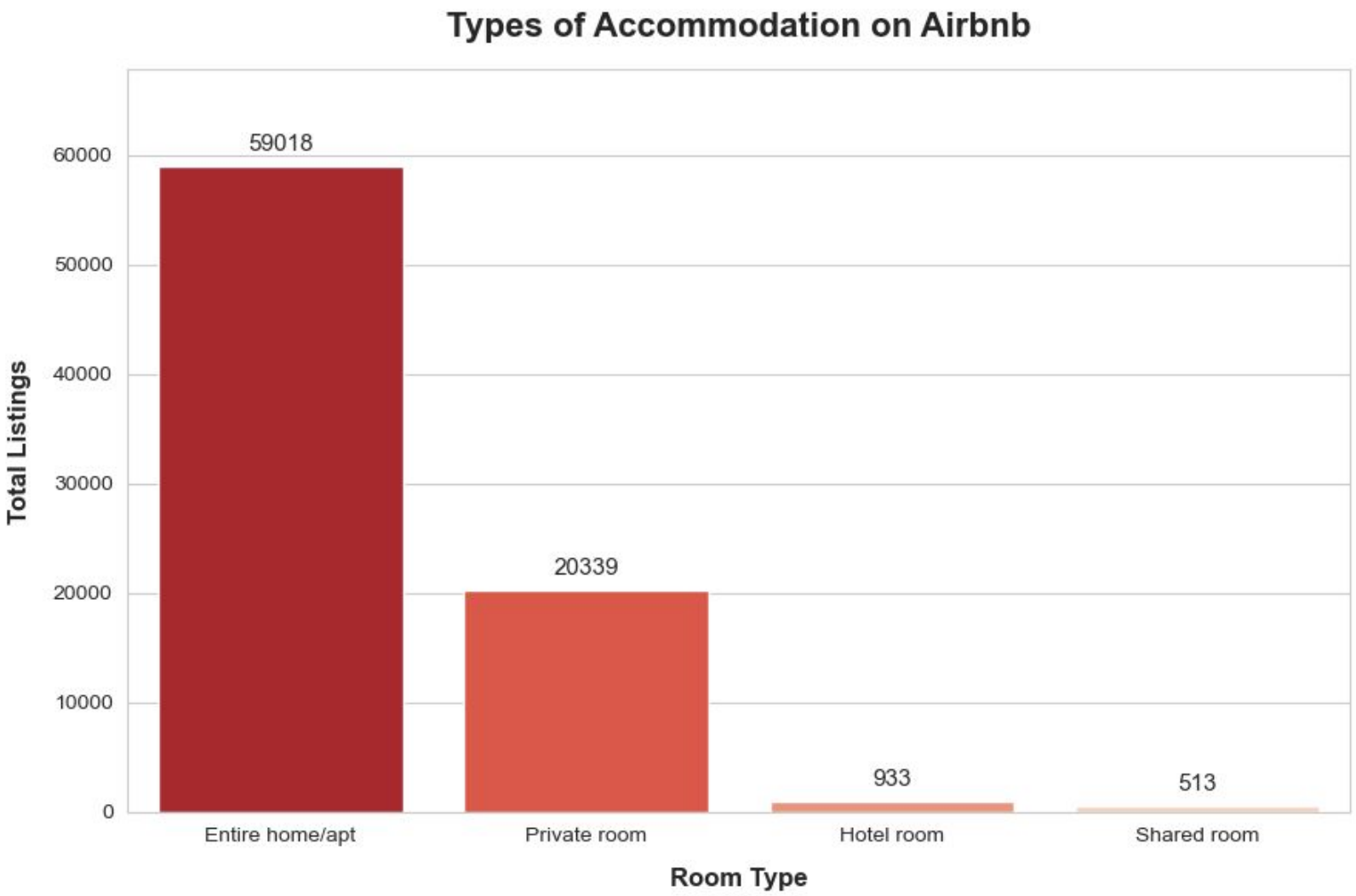
Ready for Modeling: Final dataset contains clean, numeric features suitable for Linear Regression and Clustering algorithms.

Exploratory Data Analysis

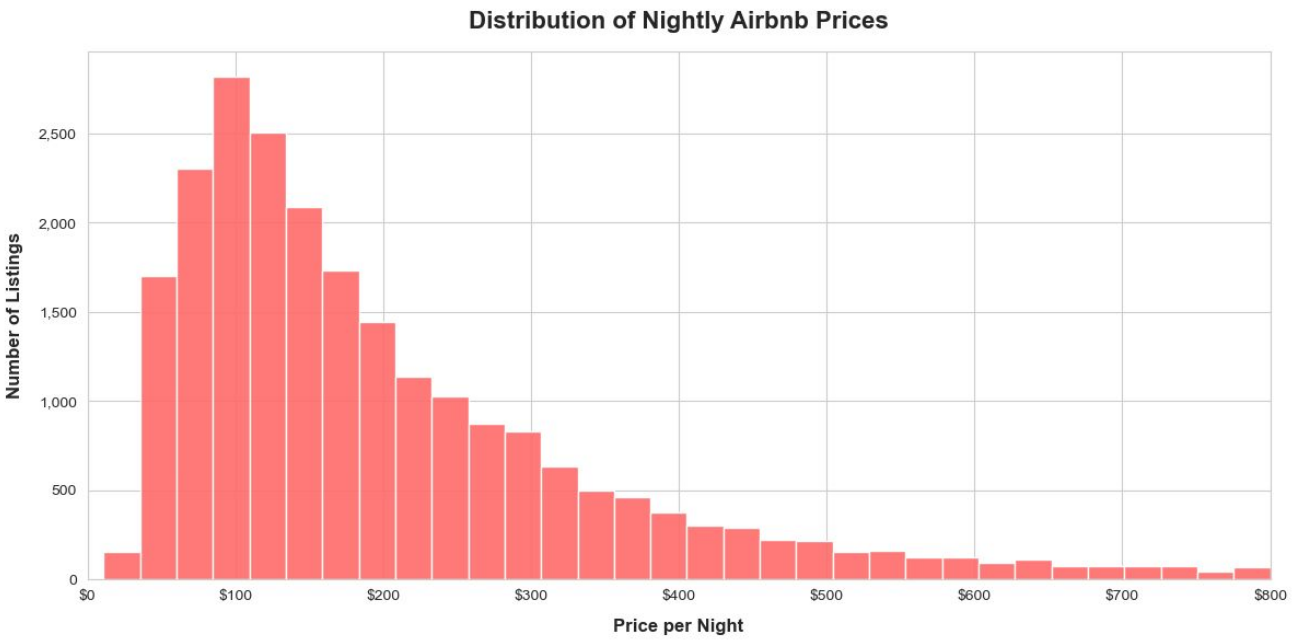
Number of Listings



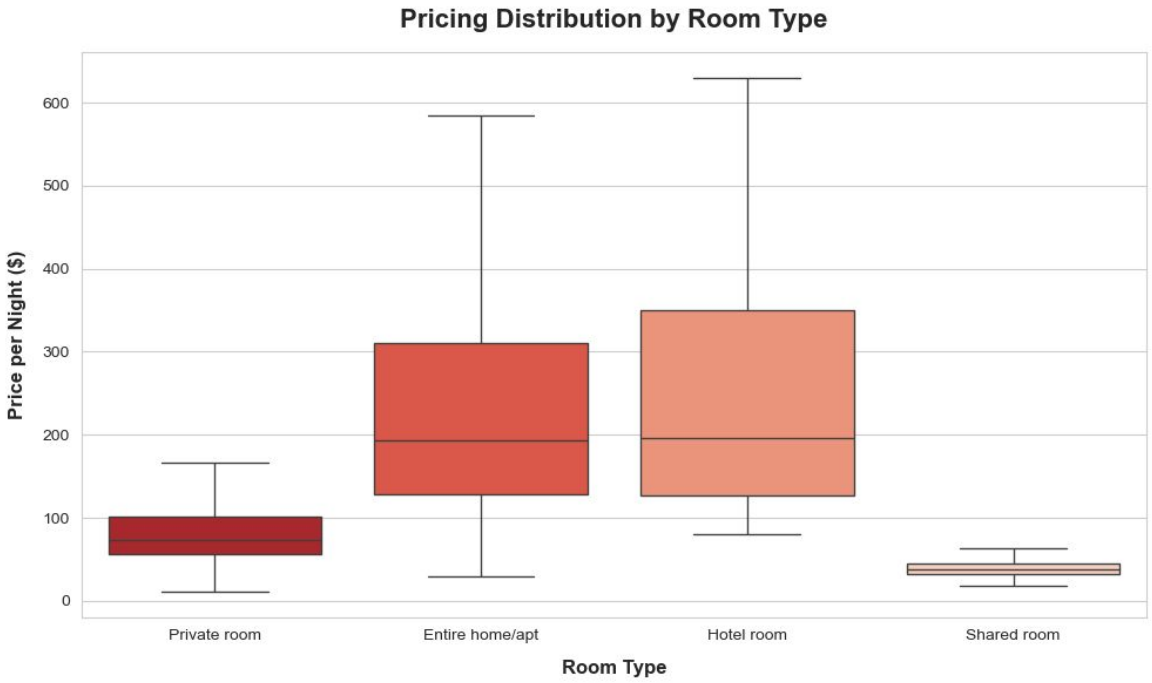
Accommodation Type



Price Distribution per Night



Price Distribution by Room Type



Predictive Question

Can clustering-based market segmentation improve the prediction of Airbnb listing demand across California?

Demand Driver Analysis

Q1 How do the key drivers of Airbnb listing demand vary across different locations in California?

Linear Regression | Random Forest | XGBoost

Segmentation Analysis

Q2 Can Airbnb listings be segmented into different market groups?

K-Means | DBSCAN

Q1 How do the key drivers of Airbnb listing demand vary across different California markets?

Feature Set (X)

Property Characteristics: Bedrooms, Bathrooms, Accommodates, Property Type

Host Attributes: Host quality such as: Host is superhost, Host Response Time, Host Identity Verified

Economic Factors: Price, Availability

Geographic Information: City, Latitude, Longitude

Target Variable (Y)

Demand (Reviews per month)

Why is 'Reviews per month' our target variable?

Since Airbnb does not publicly disclose the exact number of bookings or revenue for a listing, the frequency of reviews is the industry-standard metric to estimate how often a property is being stayed in.

Model Training Methodology

[80:20 Train/Test split]

Model 1:
Linear Regression

- Predict Airbnb demand using ***reviews per month*** as a proxy for listing performance

Model 2:
Random Forest

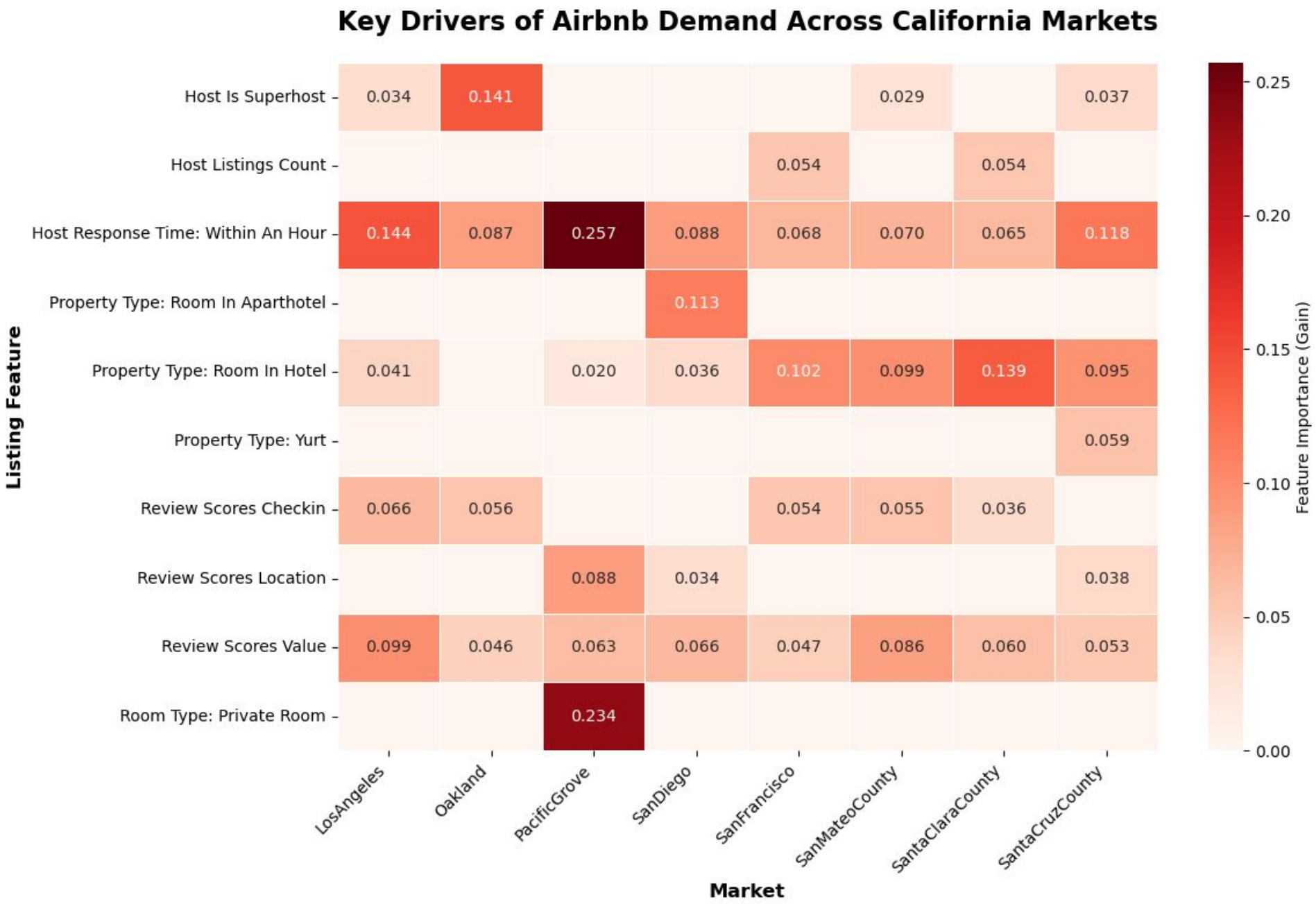
- Capture **nonlinear relationships** between listing features and demand

Model 3:
XGBoost Regressor

- Improve **prediction accuracy** by using a gradient boosting model that builds trees sequentially

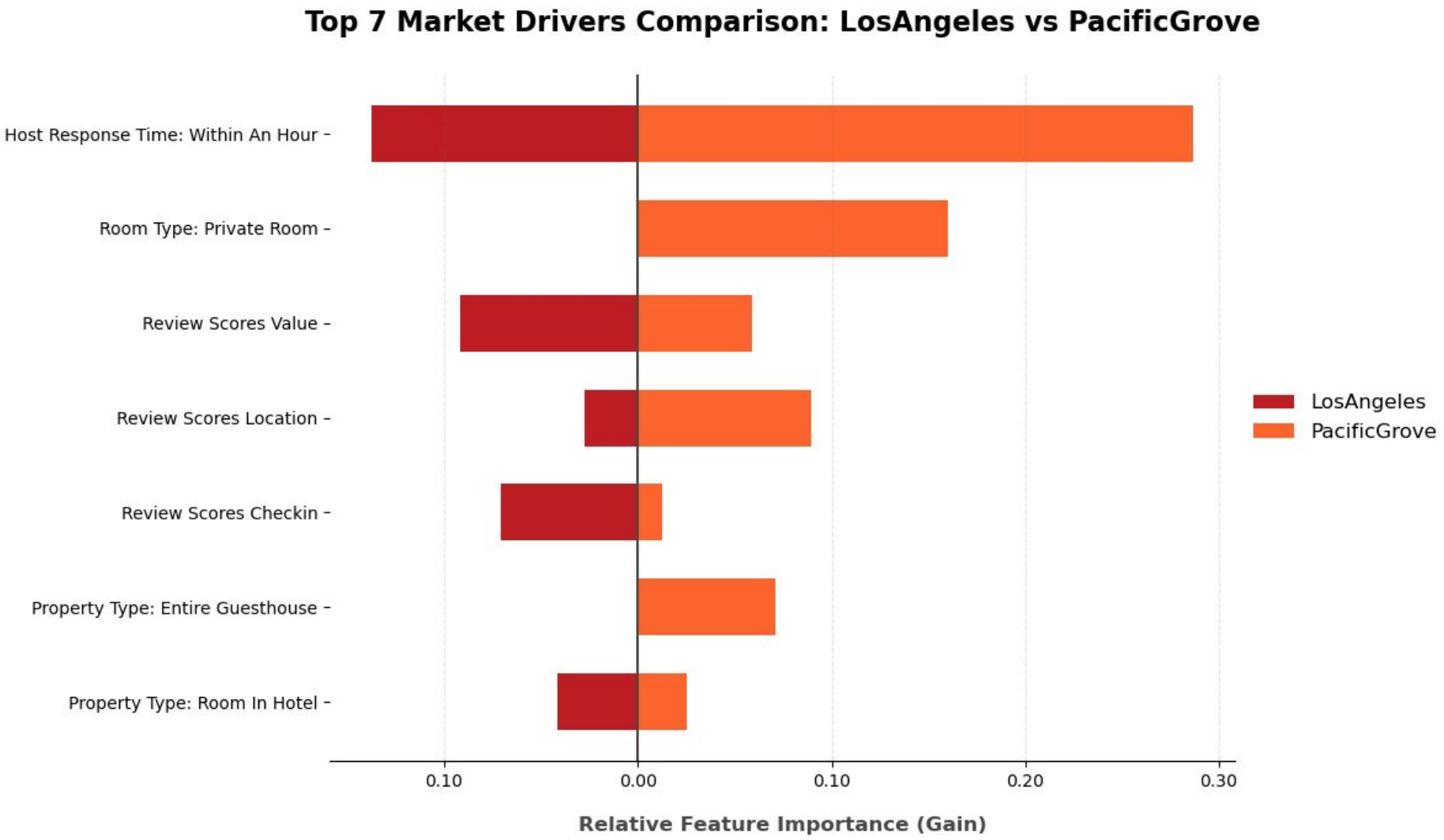
Model Performance Results

Model	R^2 Score	RMSE	MAE
Linear Regression	0.2642	1.4926	0.9102
Random Forest	0.4961	1.2352	0.5992
★ **XGBoost**	**0.5275**	**1.1961**	**0.6051**

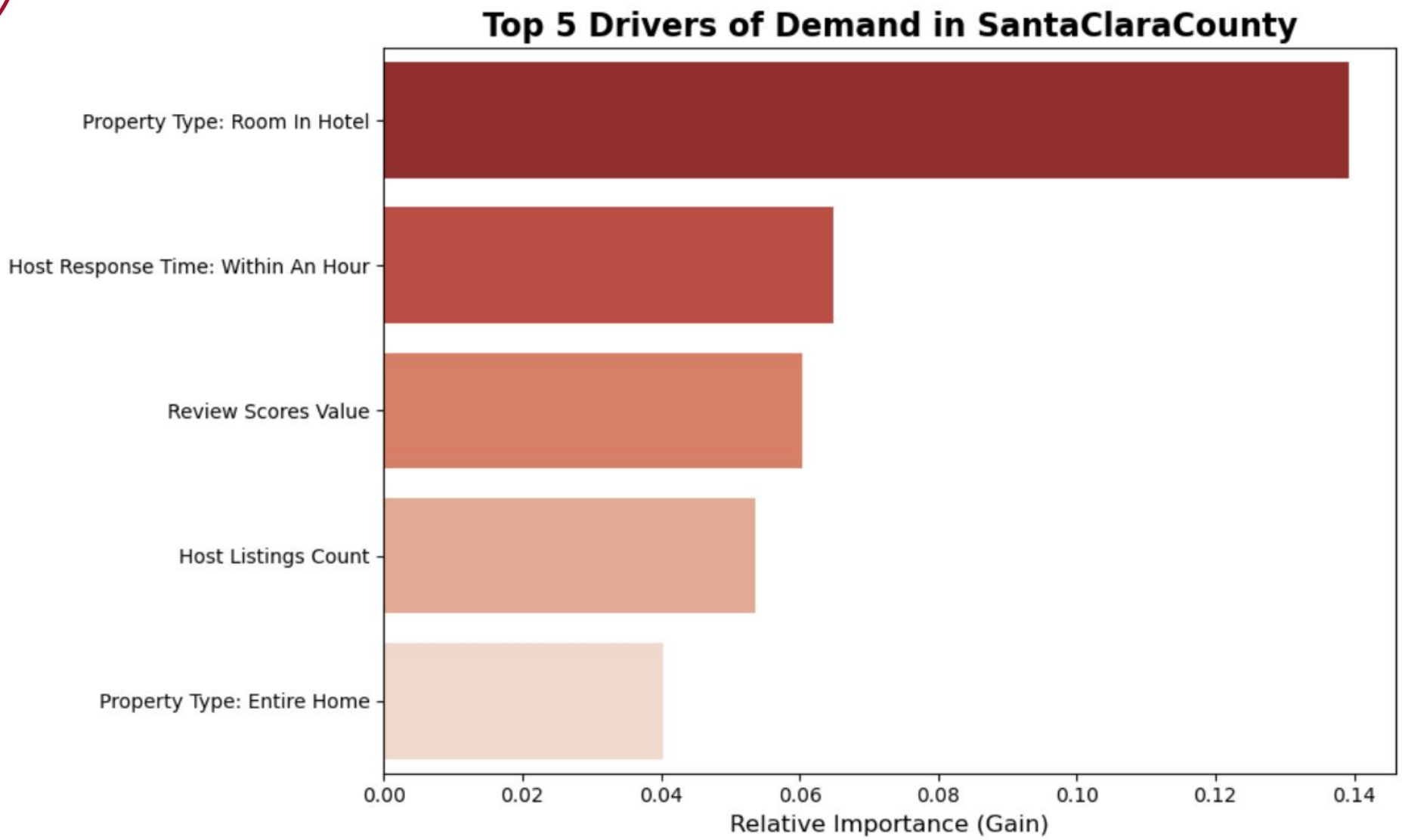


*** Feature importance in XGBoost is measured by Gain ***

Demand Drivers in a Large City vs Holiday Destination



Top Drivers of Demand in our County!



Q1 Actionable Recommendations

- **Treat communication as a core utility**, not a customer service task
  - The first to respond often wins the booking, regardless of location
  - Use AI-driven auto-responders
- The California traveler is currently showing a strong preference for predictability and professional standards
  - **Standardize Amenities:** High-end linens, branded toiletries and curated guides
  - **Emphasize privacy features:** Keyless entry and private patios
- In California, if a guest feels they overpaid relative to the quality, the value score will drop
  - **Introduce Dynamic Pricing:** Pricing must fluctuate based on local events (like Coachella or tech conferences) but never at the cost of the Value score

Q2 Can Airbnb listings be segmented into distinct market groups based on property characteristics, demand, and host behavior?

Model 1:
K-Means Clustering

- **Elbow Method** and **Silhouette method**
- Identify **optimal number of clusters**

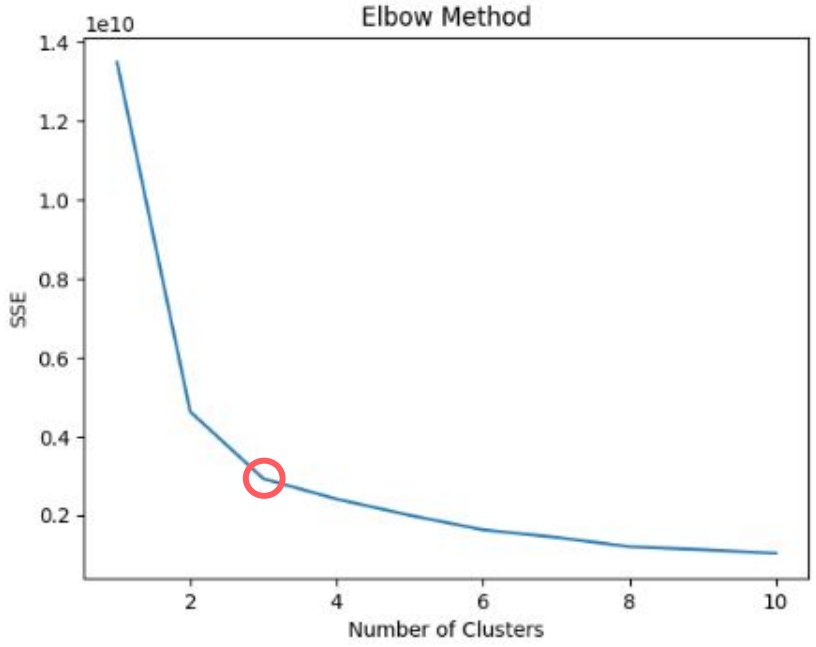
Model 2:
DBSCAN

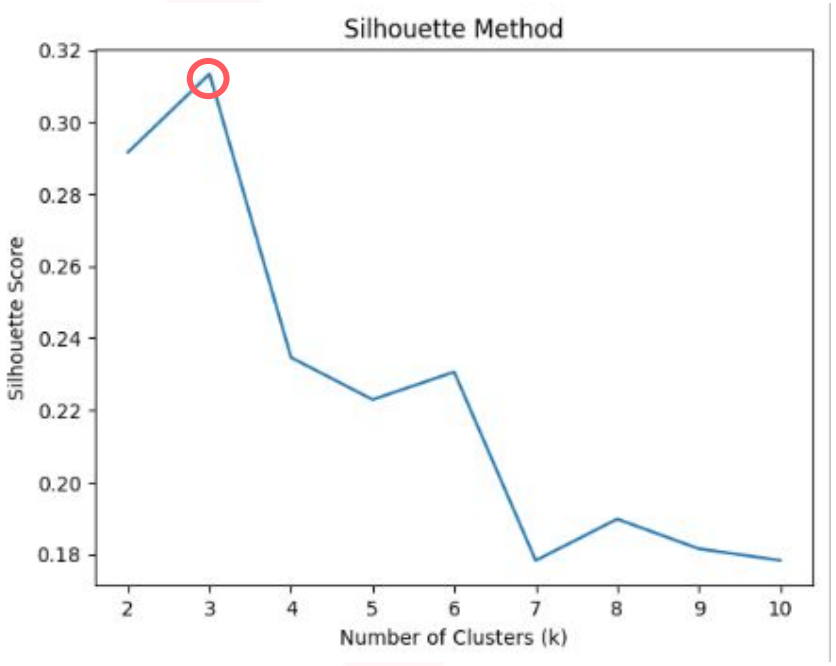
- Identify **noise and outliers**
- Detect **irregular shaped clusters**

Elbow Method vs. Silhouette Method

Selected Cluster Features:

- Property Characteristics
- Demand
- Host behaviors



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Optimal k: 3
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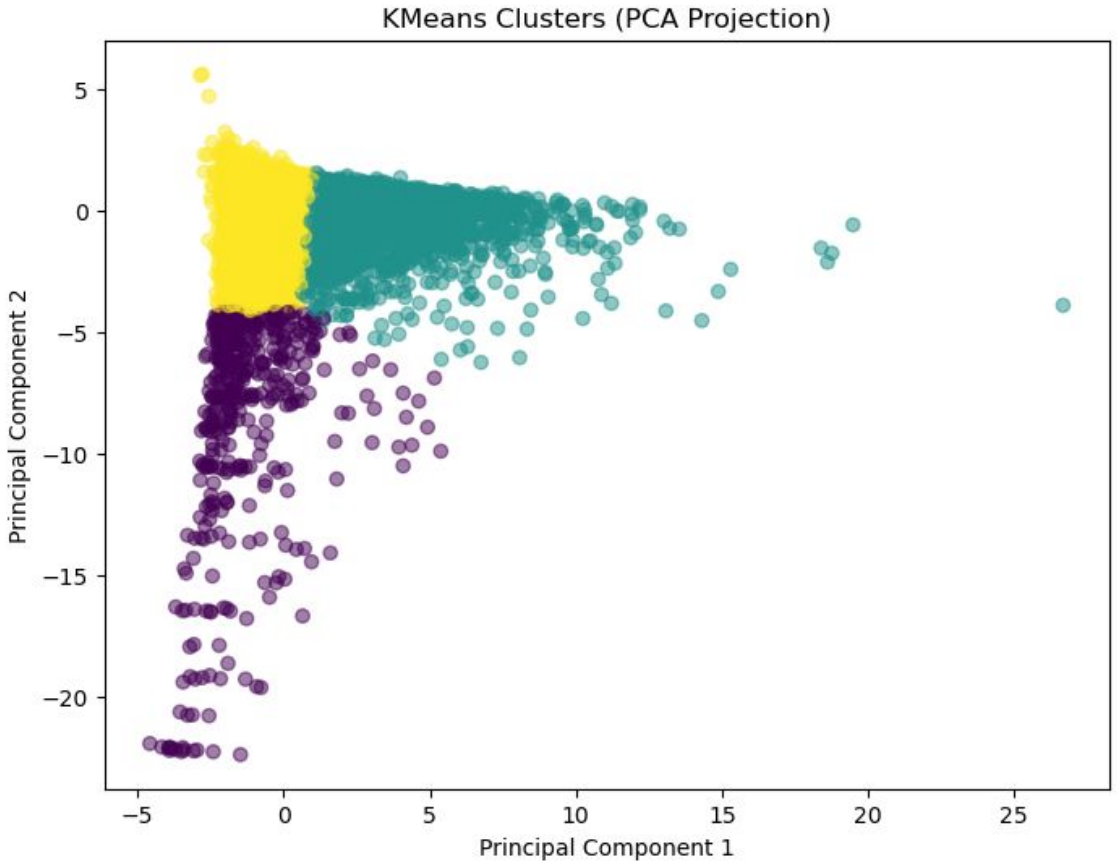
Both methods show an optimal K of 3

Results and Interpretation



- K-Means used 14 features to create clusters
- PCA can show
  - One Airbnb listing
  - Identify Cluster (color)
  - Compressed representation of 14 features (position)

Cluster Profiles

1164
2% Total Listings

Cluster 0: Commercial Hosts

Price	Accommodates	Bedroom/Bathroom	Availability_30	Rating Scores	Host Listings
161.85	3.58	1.5 / 1.4	20.57	2.94	356.63

38120
74% Total Listings

Cluster 1: Typical / Mainstream

Price	Accommodates	Bedroom/Bathroom	Availability_30	Rating Scores	Host Listings
147.82	2.95	1.2 / 1.2	14.62	4.83	57.44

12167
24% Total Listings

Cluster 2: Luxurious / Premium

Price	Accommodates	Bedroom/Bathroom	Availability_30	Rating Scores	Host Listings
458.82	8.27	3.5 / 2.7	15.68	4.85	32.90

Why DBSCAN?

DBSCAN groups observations based on data density rather than distance to centroids, enabling detection of noise points (outliers) that don't belong to any cluster.

Particularly useful for identifying **luxury homes** and **atypical properties**

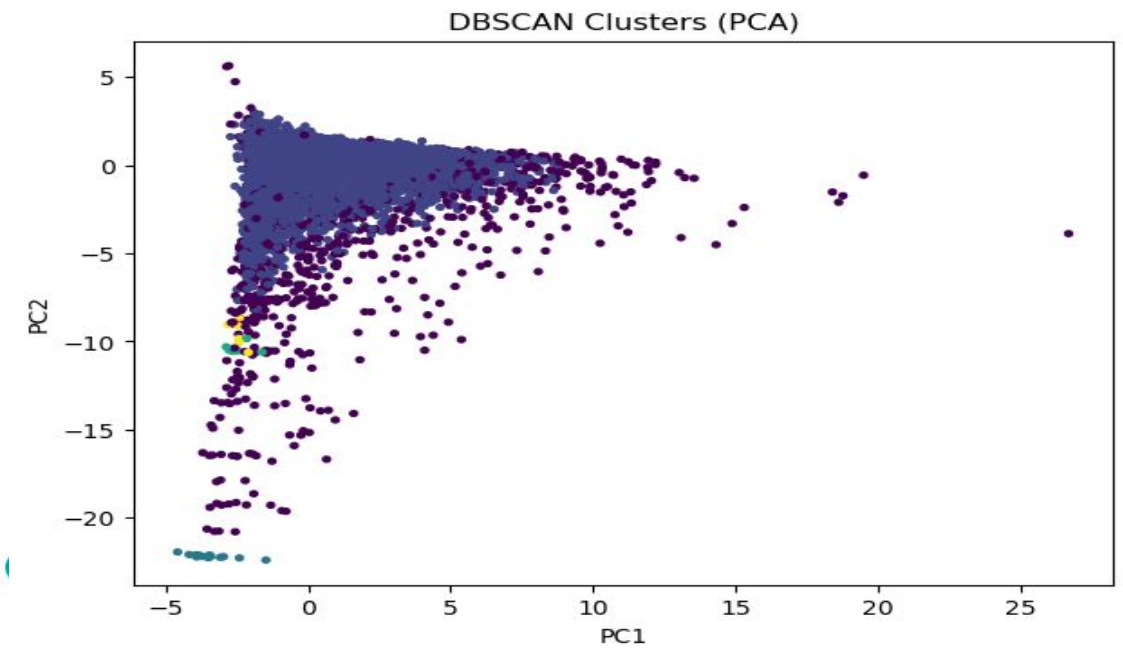
Parameter Selection

Used **k-distance plot** to determine optimal eps value, identifying the natural threshold where distance increases rapidly.

8	2.5%
Clusters Identified	Noise Points

PCA Visualization

Silhouette Score: 0.39



Total Clusters	Silhouette	Outliers
8	0.39	2.5%

Interpretation:

Compared with DBSCAN, K-Means provides clearer and more balanced market segmentation for Airbnb listings, making it more suitable for identifying meaningful clusters that improve demand prediction.

Q2 Actionable Recommendations

Recommendations for Hosts:

- ➔ Look into reasons for low ratings (Cleanliness, Value, Location, Communications)
- ➔ Improve quality and guest experience by standardizing cleaning procedures, faster customer service response time, better property maintenance
- ➔ Use dynamic pricing by adjusting prices based on seasons, holidays, weekday vs. weekend

Recommendations for Airbnb Renters:

- ➔ Choose Airbnbs from the Mainstream cluster as they tend to have lower prices, better quality, and higher ratings

Predictive Question:

Can clustering-based market segmentation improve the prediction of Airbnb listing demand across California?

Workflow: Mapping & Training

Step 1:
Data Alignment

- Mapped our K-Means cluster labels back to the original dataset

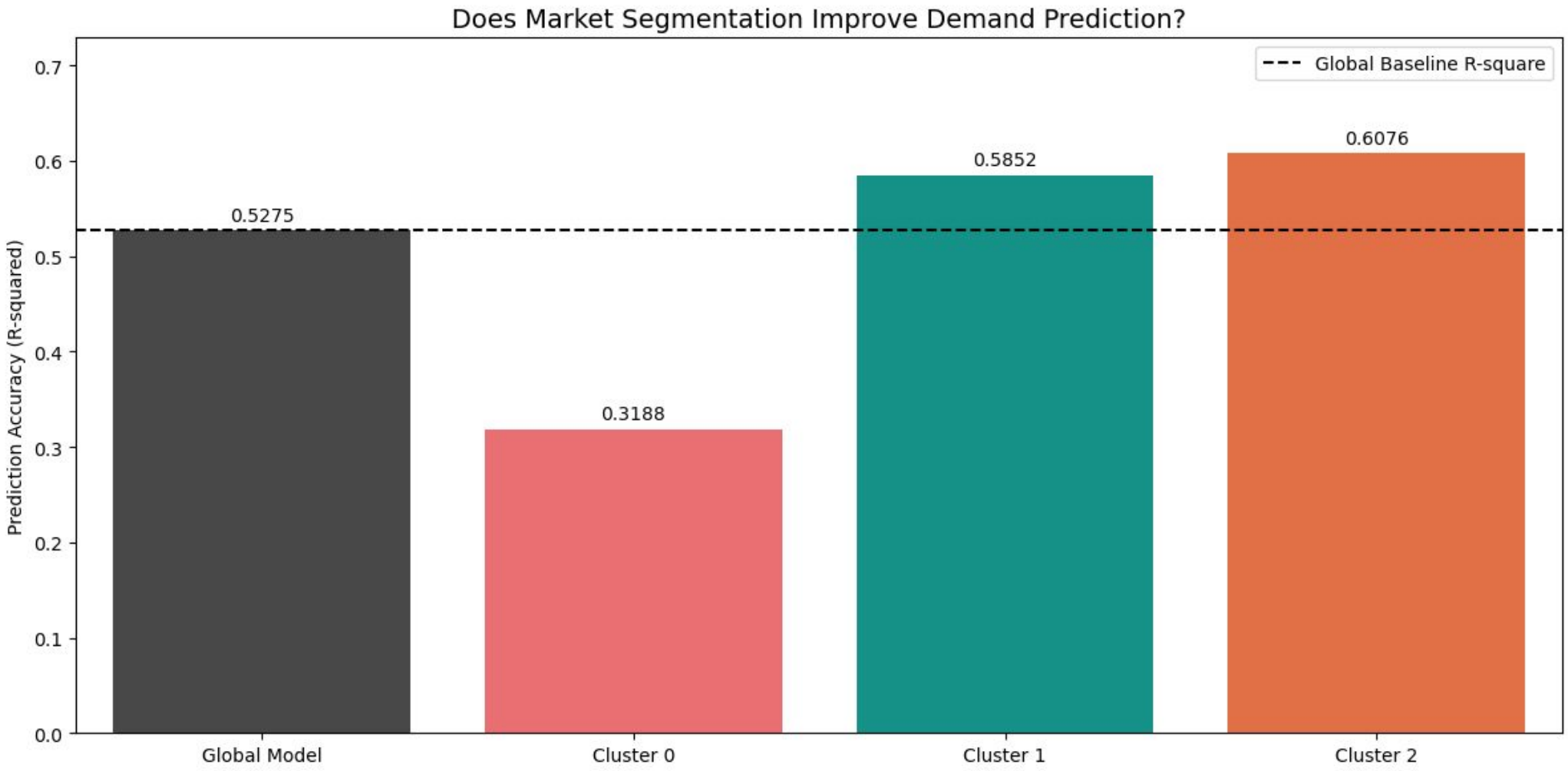
Step 2:
Training Clusters

- Trained 3 XGBoost models (1 for each cluster)

Step 3:
The Benchmark

- Compared trained clusters against the Global Baseline (R^2 = 0.5275)

Segmentation Boosts Predictive Power! Somewhat..



Conclusion

- Segmenting the California market into clusters increases demand prediction accuracy by up to **15%** for premium and mainstream listings

- Professional/Niche listings are unpredictable using standard metrics, suggesting they may rely on external marketing or corporate contracts not captured in Airbnb's public data

Thank You!

Any questions?